C14 Extracting Implied Constraints from BTOR2 Models

Implement a method for extracting word-level constraints implied by the safety properties or structure of BTOR2 hardware models. Preserve bit-vector and array expressions and formally validate every implication before use.

SUGGESTED STEPS

1. Study the implied-constraint extraction ideas in Speeding up Model Checking by Exploiting Explicit and Hidden Verification Constraints.
2. Select single- and multi-property word-level BTOR2 benchmarks from the Hardware Model Checking Competition benchmark collections.
3. Use the Pono parser or another BTOR2 frontend and implement an extraction method that proposes property-implied bit-vector bounds, equalities, disequalities, implications, masked-bit relations, and array constraints.
4. For every candidate C derived from property P, use an SMT solver to prove P implies C by checking that P and not C is unsatisfiable. Define the corresponding implication obligation for candidates derived directly from the model.
5. Run the extractor and validator over the benchmark suite, record all accepted constraints, and optionally evaluate their reuse across properties and their effect on verification time.

EXPECTED OUTCOME

1. A BTOR2 implied-constraint extraction implementation with an independent SMT-based implication validator.
2. A per-model and per-property table reporting theory features, candidate and validated constraint counts, rejected candidates, extraction time, and validation time.
3. Evidence that every property-derived constraint satisfies P implies C, with representative bit-vector or array examples.
4. If integrated into Pono or another model checker, an evaluation of verification time, solved properties, and constraint reuse across properties.

RECOMMENDED READING

Gianpiero Cabodi, Paolo Camurati, Luz Garcia, Marco Murciano, Sergio Nocco, and Stefano Quer. Speeding up Model Checking by Exploiting Explicit and Hidden Verification Constraints, DATE 2009; and the Pono model checker.